

Human Cloning: A Scientist's Story

Dr. Samuel Wood via interview

I was extremely close with my mother all my life. She was a brilliant educator, writer and wonderful woman. Sadly, she developed **complications**¹ related to diabetes. When she lost her eyesight and most of her ability to walk, it was absolutely **horrifying**² for me. She passed away from a fall seven or eight years ago. At her **funeral**, I **swore**(**swear**) that one day I'd do something about conditions like hers.

1. Complication means i) a difficulty, or a situation that complicates or makes the existing situation more complex 困境; ii) a secondary disease 并发症;
2. Horrifying means horrible, causing horror;

Years passed and I read about the work the South Koreans had done with stem cells. In 2004 and 2005 Hwang Woo-Suk **fraudulently** reported that he had succeeded in creating human **embryonic**³ stem cells by cloning.

3. 同学们学会应用该词的衍生含义: 初期的, 萌芽期的, 雏形的 Embryonic means being in the elementary stage with potential for further development. E.g. The test system is still in its **embryonic** stage.

Back then it wasn't known it was a fraud, so it was very exciting to think that a long list of diseases could be treated.

I **founded**(**found** 为该词的**原形动词**, **to establish**) the stem cell research company Stemagen with another gentleman whose father had died of ALS. We went out for drinks one night and we started talking about our parents. We wanted to do something that would be a **legacy** for them.

For Better or Worse?

The moment we decided to start Stemagen, I read all there was to read about the various cloning efforts in the past. The cloned sheep Dolly in 1997 was very interesting, but at that stage people were not focusing on the stem cell aspect of cloning; they were focusing on the reproductive possibilities of cloning.

Human reproductive cloning is just simply wrong **ethically**⁴ from a medical standpoint and a scientific standpoint, even ignoring any religious **issues**⁵ **associated with**⁶ it. The reason is that the majority of reproductive clones in other species are actually **abnormal**, with very high miscarriage rates, very high stillbirth rates, fetal **anomalies**, death soon after birth, **et cetera**⁷.

4. Ethical 表示道德的, 但 ethically 在中文里不存在“道德地”这样的副词, 我们可以理解成 in a way that avoids bringing harm to other people, in a morally good or correct manner 即“以有道德的方式去”;

5. Issue refers to an unsettled problem, or a matter in dispute. 比如说美国研究生入学考试 GRE 的写作题就是 Issue writing, 要求同学们围绕一个论题进行分析性的写作。所以 GRE 的写作也叫 The analytical writing on an issue task 即针对论题进行的分析性写作;

6. Associate with 是一个高级的高频词, 常出现在学术阅读、文书写作、四六级考试以及国外考试等等地方。它的用法如下:

i) 表示联想, to bring things together into a relationship. E.g. I often *associate* the place *with* my childhood.

ii) 表示融合, 结合, 相关联, to combine, to connect, 原文中的 any religious issues associated with it(cloning) 属于这层含义。

7. Et cetera 是 etc. 前三个字母的全称, 表示“等, 等等”。这是一个拉丁词, 是正式用语, 比 and so on 要正式得多。有不少期刊, 如 *The Economist*, *Oceania*, 都规定学术写作不得使用像“and so on”这类非正式语。

It would just be absolutely wrong to take a human being and put them through what may well involve significant⁸ suffering for really no good end. Even though people could take the techniques that we've developed and attempt to do it (or perhaps even be successful doing it), we hope that they would not.

8. Here significant means a large amount.

On the other hand, therapeutic cloning does not involve any type of risk to human life and actually provides tremendous potential for the relief⁹ of suffering in real human beings who are going through some awful(terrible, scary) things.

9. Relief means the removal(移除) of painful, distressful or oppressive feeling. The verb form of relief is relieve.

I'm a pure scientist in some ways, and I know that many different studies or findings could be used for evil. Our job as scientists is to make the most of this technology and make it available(可操作的, 可实践的, 可及的 = replicable, reproduced, replicated 可被复制的) to the greatest number of other scientists who can help us do good things with it. There's really no effective way for an individual scientist to stop someone else from using the knowledge for something they shouldn't.

We need to be honest about the techniques that we used. They need to be able to be replicated¹⁰ by other people, and so, we are providing a roadmap. I would hope that the legislation that's in place¹¹ and the great public disapproval that would result from any attempt to clone a human would dissuade anyone from going down that path.

10. Replicate means to duplicate, or repeat.

11. In place means established, 字面上来看, 表示“到位的”, 既然到位, 也就表示“已制定的”, “建好的”。原句= I would hope that **the established law** and **the great public objection** would prevent every one from doing the experiment of human reproductive cloning.

What is it they say? There is no technology that hasn't been used for some **evil**(=wicked 邪恶的) purpose at some point. Quite honestly I do think that someone will attempt human reproductive cloning. I do think it's **inevitable**(=unavoidable), and it's **virtually**(=really) impossible to legislate that away.

Claim to Fame

I am spoken of as the first man to "clone himself." There are different types of cloning. At the **cellular**(adj. of “cell”; “细胞”的形容词形式) level, yes, it's true I am the first man to clone himself. We thought a great deal about how to deal with the issue of whose cells we should use and whether we should let the world and the scientific community know who the first cellular clone was.

In the end we decided that we wanted to put a human face on cloning.

I didn't **anticipate**(expect 我没有料到) it would create the **firestorm** of **controversy**¹² that it's created, but I'm still glad we went down that path. We received thousands of e-mails and phone calls from people who need help.

12. Firestorm means a violent **outburst**. Controversy refers to the opposing voice, the dispute. 反对之声;

I think by **coming forward**¹³ and *putting a face to it*¹⁴ we made it very real, and now people around the world know that cloning is here. I believe that very soon it will be used therapeutically, so I think our purpose was served.

13. Come forward means to **become publicly known, or to provide information voluntarily** 表示主动公开;

14. Put a face to/on sth means to make something easier to understand by connecting it to real person. 所以这个句子可以理解成: I think by voluntarily making our experiment known to the public and by using my own skin cell to do the experiment, we let the world know that this therapeutic cloning did happen in our lab. It was not a fraud. It indeed happened.

Pure Science

What happens is an **informed**(已知情的) and **consenting**(同意参与实验的) woman **donates** an egg and we **remove** her genetic material from the egg. Then we place a single skin cell inside that egg.

What we're really interested in is creating disease-specific and person-specific stem cell lines. The procedure¹⁵ of taking cells from a person takes no more than a minute or two. You can take some skin cells from the arm, for example, and in one to two minutes, you can get the cells that you need to carry out(=enforce, implement) this process.

15. The difference between a process and a procedure is that a process consists of a series of procedures.

This process enables us to study the causes of specific diseases, such as Alzheimer's Disease, ALS or Parkinson's Disease, and then research a variety of treatments for these diseases. If the stem cell lines are created for any given individual and are later transplanted (移植) back into the individual, they will not be rejected(排斥) by the individual.

Sweet Success

I always thought that when our research was successful I would just be pleased that we had accomplished this when others had not. In reality, it is transcendent -- when you look through the microscope, you see what you may have looked like a long time ago, at least in part.

When I looked down and saw that cloned blastocyst, it brought tears to my eyes. I had done this for my mother, and I realized, had she only been able to live a few years longer, maybe we could have used this technology to help her. It was emotional(情绪激动的, agitated) to see that potential, which she never had a chance to experience.

There's a big misconception(=an inaccurate understanding or interpretation) out there that we decided to destroy these embryos for some reason. There was so much skepticism¹⁶ about this process because of the scientific fraud(实验造假, food fraud 食物造假) from the past that it was critical(=important) that there be no doubt that they were clones.

16. Skeptic, skepticism are the American spelling; sceptic, scepticism are the British spelling. 含义为“质疑之声”。

In the process of analysis, the embryos were destroyed by necessity(=out of necessity). In other words, to get the genetic material from inside the cells to analyze it, you have to destroy the cell. We would have loved to have been able to avoid destroying them.

Now we're working full-time on creating stem cell lines, and people are watching with great interest.

The Pope and the President

There are a variety of opponents to our work.

We were condemned¹⁷ by the Vatican and mentioned in a negative light(=mentioned

negatively) in President Bush's State of the Union address. In a sense it's an honor because it shows that we're doing something significant. It's not every day that you get condemned by the Vatican and President Bush in the same week.

17. condemn 是一个性质较严重的词汇, it means to denounce publicly, to express a very strong disapproval in public 公开、强烈谴责; condemn also means to sentence sb to death. 宣判死刑 E.g. The bribe-takers were all condemned to death.

There's usually no dialogue between the researchers in the embryonic stem cell field and those who **oppose** it.

It doesn't make sense to me that it's such an **emotional**(情绪化的) and **contentious**(受争议的) topic. **Logically**, this is not life. I agree it's a **potential** life, but the vast majority of embryos never become life. The majority **generate**(=come into existence, produce 产生), don't **implant**(植入, 移植, insert) and die. A fetus is a life. That argument makes sense to me, but it doesn't make sense to me to look at an embryo in a lab and give it all the rights of a human life.